MEMORIA DE TRABAJO: ANÁLISIS DE MEMORIA RAM

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1. Análisis Inicial de la Memoria

Al analizar la memoria con Volatility, notamos que **no existe un perfil predefinido** para la máquina. Sin embargo, ejecutando:

```
volatility -f RAM.bin banners.Banners
```

Obtenemos el siguiente resultado:

```
┌──(kali㉿kali)-[~]
└─$ vol -f RAM.bin banners.Banners
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
Offset  Banner

0x1a000e0       Linux version 4.4.0-1061-aws (buildd@lgw01-amd64-024) (gcc version 5.4.0 20160609 (Ubuntu 5.4.0-6ubuntu1~16.04.9) ) #70-Ubuntu SMP Fri May 25 21:47:34 UTC 2018 (Ubuntu 4.4.0-1061.70-aws 4.4.131)
0x210c2e4       Linux version 4.4.0-1061-aws (buildd@lgw01-amd64-024) (gcc version 5.4.0 20160609 (Ubuntu 5.4.0-6ubuntu1~16.04.9) ) #70-Ubuntu SMP Fri May 25 21:47:34 UTC 2018 (Ubuntu 4.4.0-1061.70-aws 4.4.131)
0x3a6f3090      Linux version 4.4.0-1061-aws (buildd@lgw01-amd64-024) (gcc version 5.4.0 20160609 (Ubuntu 5.4.0-6ubuntu1~16.04.9) ) #70-Ubuntu SMP Fri May 25 21:47:34 UTC 2018 (Ubuntu 4.4.0-1061.70-aws 4.4.131)
0x3fafe0d0      Linux version 4.4.0-1061-aws (buildd@lgw01-amd64-024) (gcc version 5.4.0 20160609 (Ubuntu 5.4.0-6ubuntu1~16.04.9) ) #70-Ubuntu SMP Fri May 25 21:47:34 UTC 2018 (Ubuntu 4.4.0-1061.70-aws 4.4.131)
```

El resultado indica:

- **Kernel:** 4.4.0-1061-aws
- **Sistema Operativo:** Ubuntu 16.04

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2. Descarga y Extracción de Símbolos del Kernel

Para continuar, necesitamos los símbolos del kernel. Realiza los siguientes pasos:

1. Ve a la página de Kernel Ubuntu AWS
2. Descarga el archivo:
   - `linux-image-4.4.0-1061-aws_4.4.0-1061.64_amd64.deb`
3. Extrae el contenido:

   ```
   dpkg -x linux-image-4.4.0-1061-aws_4.4.0-1061.64_amd64.deb .
   ```

4. Obtén el archivo `vmlinux-4.4.0-1061-aws` (contiene los símbolos del kernel).

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3. Generación de Símbolos con dwarf2json

Para convertir el archivo `vmlinux` a un formato compatible con Volatility:

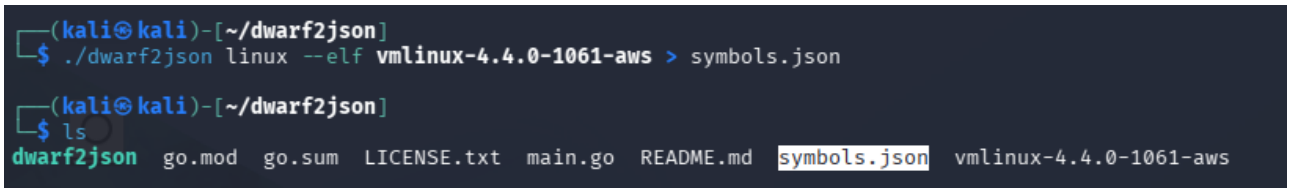
1. Clona y compila `dwarf2json`:

   ```
   git clone https://github.com/volatilityfoundation/dwarf2json
   cd dwarf2json
   ```

```
go build
```

2. Genera el archivo de símbolos:

```
./dwarf2json linux --elf vmlinux-4.4.0-1061-aws > symbols.json
```



3. Organiza los símbolos:

```
mkdir symbols
mv symbols.json symbols/
```

---

4. Análisis de Procesos con Volatility

Con los símbolos generados, ejecuta:

```
vol -f RAM.bin --symbol-dirs=./symbols linux.pslist.PsList
```

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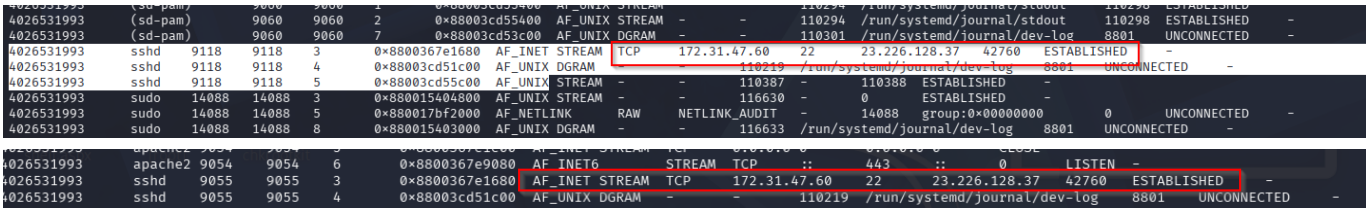
5. Hallazgos Relevantes en la Memoria RAM

A continuación se presentan las principales evidencias encontradas durante el análisis de la memoria RAM, acompañadas de sus respectivas capturas y una breve descripción profesional de cada hallazgo:

5.1 Conexiones SSH desde IP desconocida



Ambas capturas muestran conexiones SSH entrantes desde una dirección IP no reconocida, lo que sugiere un posible acceso no autorizado al sistema. Es fundamental investigar la procedencia de esta IP y determinar si corresponde a una actividad legítima o maliciosa.

5.2 Historial de comandos sospechoso

```
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo apt install apache2 libapache2-mod-php php-mysql
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo apt upgrade
    9126 bash                  2018-07-24 05:24:19 UTC+0000   mysql -uadmin -p -hganga.ctmbcxcdb3us.eu-central-1.rds.
amazonaws.com ganga
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo service apache2 restart
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo apt update
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo letsencrypt --apache -d ganga.site -d www.ganga.si
te
    9126 bash                  2018-07-24 05:24:19 UTC+0000   apt-cache search certbot
    9126 bash                  2018-07-24 05:24:19 UTC+0000   mysql -uadmin -p -hganga.ctmbcxcdb3us.eu-central-1.rds.
amazonaws.com ganga
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo letsencrypt --apache -d ganga.site -d www.ganga.si
te
    9126 bash                  2018-07-24 05:24:19 UTC+0000   cd /etc/apache2
    9126 bash                  2018-07-24 05:24:19 UTC+0000   certbot
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo service apache restart
    9126 bash                  2018-07-24 05:24:19 UTC+0000   ls -l ../mods-enabled/
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi 000-default.conf
    9126 bash                  2018-07-24 05:24:19 UTC+0000   ls -l
    9126 bash                  2018-07-24 05:24:19 UTC+0000   cd ..
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi /etc/apache2/sites-enabled/000-default-le-ssl.c
onf
    9126 bash                  2018-07-24 05:24:19 UTC+0000   tail /var/log/apache2/access.log
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi ../apache2.conf
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi /etc/apache2/sites-enabled/000-default-le-ssl.c
onf
    9126 bash                  2018-07-24 05:24:19 UTC+0000   letsencrypt
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo apt-get install python-certbot-apache
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo rm index.html
    9126 bash                  2018-07-24 05:24:19 UTC+0000   ls -l ~
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo letsencrypt --authenticator webroot --installer ap
ache --webroot-path /var/www/html -d ganga.site -d www.ganga.site
    9126 bash                  2018-07-24 05:24:19 UTC+0000   ls -l ..
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi /etc/apache2/sites-enabled/000-default-le-ssl.c
onf
    9126 bash                  2018-07-24 05:24:19 UTC+0000   find /etc/apche2 -type f -exec grep -i Logformat {} \;
    9126 bash                  2018-07-24 05:24:19 UTC+0000   mysql -u ganga -p -h ganga.ctmbcxcdb3us.eu-central-1.rd
s.amazonaws.com
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi /etc/apache2/sites-enabled/000-default-le-ssl.c
onf
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo service apache2 restart
    9126 bash                  2018-07-24 05:24:19 UTC+0000   find /etc/apache2 -type f -exec grep -Hi Logformat {} \
;
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo apt-get install python-letsencrypt-apache
    9126 bash                  2018-07-24 05:24:19 UTC+0000   tail /var/log/apache2/access.log
    9126 bash                  2018-07-24 05:24:19 UTC+0000   find /etc/apache2 -type f -exec grep -i Logformat {} \;
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo service apache2 restart
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi /etc/apache2/sites-enabled/000-default-le-ssl.c
onf
    9126 bash                  2018-07-24 05:24:19 UTC+0000   mysql -u admin -p -h ganga.ctmbcxcdb3us.eu-central-1.rd
s.amazonaws.com
    9126 bash                  2018-07-24 05:24:19 UTC+0000   sudo vi /etc/apache2/sites-enabled/000-default-le-ssl.c
```

En esta evidencia se observa el historial de comandos ejecutados en la máquina. Se infiere que se realizaron modificaciones en el sitio web alojado, lo que podría indicar un intento de alteración o compromiso de la integridad del servicio.

5.3 Uso de Yarascan en Apache y SSH

```
┌──(vola)─(kali㉿kali)-[~/Desktop/Volatilty/volatility]
└─$ python2 vol.py -f /home/kali/Desktop/RAM.bin --profile=LinuxUbuntu_4_4_0-1061-aws_4_4_0-1061_70_amd64x64 linux_yarascan -Y "23.226.128.37"
Volatility Foundation Volatility Framework 2.6.1
Task: systemd-journal pid 413 rule r1 addr 0x7fd18c5f7bdb
0x7fd18c5f7bdb  32 33 2e 32 32 36 2e 31 32 38 2e 33 37 20 70 6f   23.226.128.37.po
0x7fd18c5f7beb  72 74 20 34 32 37 36 30 20 73 73 68 32 3a 20 52   rt.42760.ssh2:.R
0x7fd18c5f7bfb  53 41 20 53 48 41 32 35 36 3a 51 32 37 70 57 36   SA.SHA256:Q27pW6
0x7fd18c5f7c0b  64 44 59 50 4a 38 4e 30 6d 42 58 36 4c 38 53 4f   dDYPJ8N0mBX6L8SO
0x7fd18c5f7c1b  38 4f 51 37 4c 56 53 64 4e 64 6d 31 78 78 7a 79   8OQ7LVSdNdm1xxzy
0x7fd18c5f7c2b  42 54 32 33 59 01 00 00 00 00 00 00 00 49 00 00   BT23Y........I..
0x7fd18c5f7c3b  00 00 00 00 00 75 20 1a 4a 20 6d d3 d0 00 00 00   .....u..J.m.....
0x7fd18c5f7c4b  00 00 00 00 00 80 f3 02 00 00 00 00 00 50 fd 02   .............P..
0x7fd18c5f7c5b  00 00 00 00 00 58 02 03 00 00 00 00 00 02 00 00   .....X..........
0x7fd18c5f7c6b  00 00 00 00 00 5f 50 49 44 3d 39 30 35 35 00 00   ....._PID=9055..
0x7fd18c5f7c7b  00 00 00 00 00 01 00 00 00 00 00 00 00 5d 00 00   .............]..
0x7fd18c5f7c8b  00 00 00 00 00 df 95 bc 4d 3d a4 5b c6 00 00 00   ........M=.[....
0x7fd18c5f7c9b  00 00 00 00 00 08 ba 01 00 00 00 00 00 50 fd 02   .............P..
0x7fd18c5f7cab  00 00 00 00 00 90 02 03 00 00 00 00 00 02 00 00   ................
0x7fd18c5f7cbb  00 00 00 00 00 5f 43 4d 44 4c 49 4e 45 3d 73 73   ....._CMDLINE=ss
0x7fd18c5f7ccb  68 64 3a 20 75 62 75 6e 74 75 20 5b 70 72 69 76   hd:.ubuntu.[priv
Task: dbus-daemon pid 1148 rule r1 addr 0x56113f009f54
0x56113f009f54  32 33 2e 32 32 36 2e 31 32 38 2e 33 37 00 00 00   23.226.128.37...
0x56113f009f64  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009f74  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009f84  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009f94  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009fa4  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009fb4  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009fc4  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009fd4  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x56113f009fe4  00 00 00 00 00 00 00 00 00 00 00 00 80 90 00 3f   ...............?
0x56113f009ff4  11 56 00 00 00 00 00 00 00 00 00 00 00 00 00 00   .V..............
0x56113f00a004  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
```

```
0x7f40b23d923c  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
Task: apache2 pid 27428 rule r1 addr 0x7f0663fd8898
0x7f0663fd8898  32 33 2e 32 32 36 2e 31 32 38 2e 33 37 00 00 00   23.226.128.37...
0x7f0663fd88a8  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd88b8  4e 55 4c 4c 00 66 61 76 69 63 6f 6e 2e 69 63 6f   NULL.favicon.ico
0x7f0663fd88c8  20 48 54 54 50 2f 31 2e 31 00 64 61 63 74 65 64   .HTTP/1.1.dacted
0x7f0663fd88d8  2f 65 78 70 65 72 74 2f 6d 61 69 6e 74 65 6e 61   /expert/maintena
0x7f0663fd88e8  6e 63 65 2f 64 69 61 67 6e 6f 73 74 69 63 2f 00   nce/diagnostic/.
0x7f0663fd88f8  67 61 6e 67 61 2e 73 69 74 65 3a 34 34 33 00 00   ganga.site:443..
0x7f0663fd8908  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8918  00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00   ................
0x7f0663fd8928  00 00 00 00 02 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8938  26 00 00 00 00 00 00 00 7f e1 04 00 00 00 00 00   &...............
0x7f0663fd8948  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8958  51 8e 3b ac a8 71 05 00 99 8e 3b ac a8 71 05 00   Q.;..q....;..q..
0x7f0663fd8968  0c 38 ab e5 b7 71 05 00 01 00 00 00 00 00 00 00   .8...q..........
0x7f0663fd8978  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8988  00 00 00 00 00 00 00 00 3a 3a 31 00 34 32 2e 35   ........::1.42.5
Task: apache2 pid 27428 rule r1 addr 0x7f0663fd8c78
0x7f0663fd8c78  32 33 2e 32 32 36 2e 31 32 38 2e 33 37 00 00 00   23.226.128.37...
0x7f0663fd8c88  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8c98  00 45 54 20 2f 20 48 54 54 50 2f 31 2e 31 00 2f   .ET./.HTTP/1.1./
0x7f0663fd8ca8  70 6c 75 67 69 6e 73 2f 72 65 66 6c 65 78 2d 67   plugins/reflex-g
0x7f0663fd8cb8  61 6c 6c 65 72 79 2f 72 65 61 64 6d 65 2e 74 78   allery/readme.tx
0x7f0663fd8cc8  74 20 48 54 54 50 2f 31 2e 31 00 70 6c 6f 61 00   t.HTTP/1.1.ploa.
0x7f0663fd8cd8  00 61 6e 67 61 2e 73 69 74 65 3a 38 30 00 00 00   .anga.site:80...
0x7f0663fd8ce8  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8cf8  00 00 00 00 00 00 00 00 05 00 00 00 00 00 00 00   ................
0x7f0663fd8d08  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8d18  2b 00 00 00 00 00 00 00 7b 72 04 00 00 00 00 00   +.......{r......
0x7f0663fd8d28  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8d38  ff f3 74 ab a8 71 05 00 45 f4 74 ab a8 71 05 00   ..t..q..E.t..q..
0x7f0663fd8d48  5a 3a 84 ab a8 71 05 00 0f 00 00 00 00 00 00 00   Z:...q..........
0x7f0663fd8d58  01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd8d68  00 00 00 00 00 00 00 00 3a 3a 31 00 34 32 2e 35   ........::1.42.5
Task: apache2 pid 27428 rule r1 addr 0x7f0663fd9058
0x7f0663fd9058  32 33 2e 32 32 36 2e 31 32 38 2e 33 37 00 00 00   23.226.128.37...
0x7f0663fd9068  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x7f0663fd9078  00 45 54 20 2f 20 48 54 54 50 2f 31 2e 30 00 74   .ET./.HTTP/1.0.t
0x7f0663fd9088  2f 70 6c 75 67 69 6e 73 2f 72 65 66 6c 65 78 2d   /plugins/reflex-
0x7f0663fd9098  67 61 6c 6c 65 72 79 2f 61 64 6d 69 6e 2f 73 63   gallery/admin/sc
0x7f0663fd90a8  72 69 70 74 73 2f 46 69 6c 65 55 70 6c 6f 61 00   ripts/FileUploa.
```

Las siguientes capturas evidencian conexiones desde la misma IP utilizando la herramienta Yarascan sobre los procesos de Apache y SSH. Esto puede indicar una búsqueda activa de patrones maliciosos o la presencia de un atacante analizando servicios críticos del sistema.

5.4 Detección de posible malware

```
┌──(vola)─(kali㉿kali)-[~/Desktop/Volatilty/volatility]
└─$ python2 vol.py -f /home/kali/Desktop/RAM.bin --profile=LinuxUbuntu_4_4_0-1061-aws_4_4_0-1061_70_amd64x64 linux_malfind
Volatility Foundation Volatility Framework 2.6.1

Process: apache2 Pid: 5573 Address: 0x7f0663ff3000 File: Anonymous Mapping
Protection: VM_READ|VM_WRITE|VM_EXEC
Flags: VM_READ|VM_WRITE|VM_EXEC|VM_MAYREAD|VM_MAYWRITE|VM_MAYEXEC|VM_ACCOUNT|VM_CAN_NONLINEAR

0x007f0663ff3000  b8 17 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x007f0663ff3010  53 41 57 41 56 41 55 55 48 8b df 48 81 ec 20 01   SAWAVAUUH..H....
0x007f0663ff3020  00 00 48 8b 43 10 48 83 e8 01 48 8d 74 24 38 48   ..H.C.H...H.t$8H
0x007f0663ff3030  c7 c1 11 00 00 00 48 89 46 08 48 8d 76 08 48 83   ......H.F.H.v.H.

0x7f0663ff3000 b817000000        MOV EAX, 0x17
0x7f0663ff3005 0000              ADD [RAX], AL
0x7f0663ff3007 0000              ADD [RAX], AL
0x7f0663ff3009 0000              ADD [RAX], AL
0x7f0663ff300b 0000              ADD [RAX], AL
0x7f0663ff300d 0000              ADD [RAX], AL
0x7f0663ff300f 005341            ADD [RBX+0x41], DL
0x7f0663ff3012 57                PUSH RDI
0x7f0663ff3013 4156              PUSH R14
0x7f0663ff3015 4155              PUSH R13
0x7f0663ff3017 55                PUSH RBP
0x7f0663ff3018 488bdf            MOV RBX, RDI
0x7f0663ff301b 4881ec20010000    SUB RSP, 0x120
0x7f0663ff3022 488b4310          MOV RAX, [RBX+0x10]
0x7f0663ff3026 4883e801          SUB RAX, 0x1
0x7f0663ff302a 488d742438        LEA RSI, [RSP+0x38]
0x7f0663ff302f 48c7c111000000    MOV RCX, 0x11
0x7f0663ff3036 48894608          MOV [RSI+0x8], RAX
0x7f0663ff303a 488d7608          LEA RSI, [RSI+0x8]
0x7f0663ff303e 48                DB 0x48
0x7f0663ff303f 83                DB 0x83

Process: apache2 Pid: 6262 Address: 0x7f0663fb6000 File: Anonymous Mapping
Protection: VM_READ|VM_WRITE|VM_EXEC
Flags: VM_READ|VM_WRITE|VM_EXEC|VM_MAYREAD|VM_MAYWRITE|VM_MAYEXEC|VM_ACCOUNT|VM_CAN_NONLINEAR

0x007f0663fb6000  f0 25 00 00 00 00 00 00 00 00 00 00 00 00 00 00   .%..............
0x007f0663fb6010  53 41 57 41 56 41 55 55 48 8b df 48 81 ec 80 00   SAWAVAUUH..H....
0x007f0663fb6020  00 00 48 8b 43 10 48 83 e8 01 48 89 44 24 40 48   ..H.C.H...H.D$@H
0x007f0663fb6030  89 dd 48 89 d8 48 8b 58 08 4c 8b 78 18 48 8b 08   ..H..H.X.L.x.H..

0x7f0663fb6000 f02500000000      AND EAX, 0x0
0x7f0663fb6006 0000              ADD [RAX], AL
0x7f0663fb6008 0000              ADD [RAX], AL
0x7f0663fb600a 0000              ADD [RAX], AL
0x7f0663fb600c 0000              ADD [RAX], AL
0x7f0663fb600e 0000              ADD [RAX], AL
```

La última captura corresponde a la detección de un archivo o proceso sospechoso, potencialmente asociado a malware presente en el equipo. Se recomienda realizar un análisis exhaustivo de este elemento para confirmar su naturaleza y tomar las medidas de remediación necesarias.